NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR MODEL-A & APEX, ALL IITS | DT: 09/01/2026
Time: 3 Hrs | GTM – 13 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
D	D	A	A	B	A	B	D	C	B
11	12	13	14	15	16	17	18	19	20
C	C	B	D	A	D	B	C	D	A
21	22	23	24	25					
4	400	45	1	13					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	C	D	A	C	D	B	B	C	C
36	37	38	39	40	41	42	43	44	45
C	C	C	B	D	B	A	A	C	A
46	47	48	49	50					
48	28	45	5	13					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	A	D	A	A	B	D	D	C	C
61	62	63	64	65	66	67	68	69	70
A	B	B	D	B	B	C	D	D	A
71	72	73	74	75					
2	29	26	18	0					

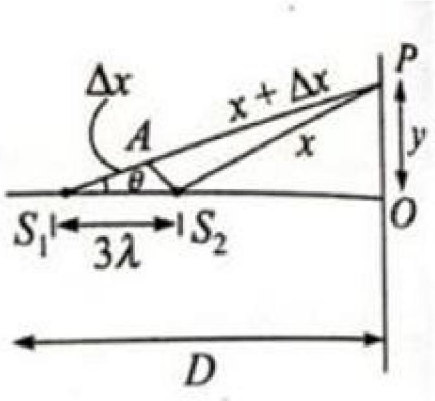
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HINTS
PHYSICS

1. $\Delta x = d\cos\theta = n\lambda$

in this case path difference is $d\cos\theta$



So, $n\lambda = 3\lambda\cos\theta \Rightarrow \cos\theta = \frac{2}{3}$

$\Rightarrow \frac{D}{\sqrt{D^2+y^2}} = \frac{2}{3} \Rightarrow 3D = \sqrt{D^2+y^2}$

$\Rightarrow y = \frac{\sqrt{5}}{2}D$

2. $i = \frac{9}{3} = \frac{V_{1\Omega}}{1} = \frac{V_{AB}}{R_{AB}} = 3A$

$V_{1\Omega} = 3\,\text{volt}$

$V_{AB} = 16 - 9 - 3 = 4V$

$\frac{9}{3} = \frac{V_{AB}}{R_{AB}}$

$\frac{9}{3} = \frac{4}{R_{AB}}$

$R_{AB} = \frac{4}{3}\Omega$

$\frac{4R}{4+R} = \frac{4}{3}$

$R = 2\Omega$

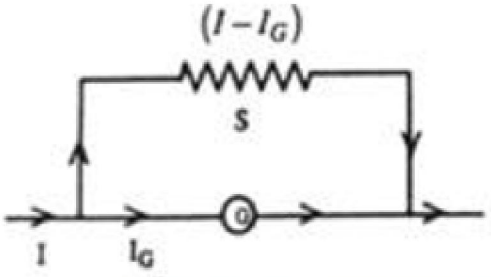
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3. The dimensions of

$$\frac{EJ^2}{M^5G^2}=\frac{\left(ML^2T^{-2}\right)^2\left(ML^2T^{-1}\right)^2}{\left(M^5\right)\left(M^{-1}L^3T^{-2}\right)^2}=1$$

So it represents dimensions quantity like angle.

4. In case of a galvanometer, $I \propto \theta$



$$n=\frac{i}{i_g}=5; s=\frac{G}{n-1}\Rightarrow 12=\frac{G}{4}\Rightarrow G=48\Omega$$

5. $$v_1=\frac{(m_1-m_2)\upsilon}{m_1+m_2}+0$$

Since, $m_1=m$ and $m_2=\frac{m}{2}$

$$\Rightarrow \upsilon=\frac{\upsilon}{3}$$

$$\Rightarrow p_1=m\left(\frac{\upsilon}{3}\right)$$

And $$\upsilon_2=\frac{2m_1\upsilon}{m_1+m_2}+0$$

$$\Rightarrow \upsilon_2=\frac{4\upsilon}{3}$$

$$\Rightarrow p_2=\frac{m}{2}\left(\frac{4\upsilon}{3}\right)=\frac{2m\upsilon}{3}$$

Since $\lambda=\frac{h}{p}$

$$=\frac{\lambda_A}{\lambda_B}=\frac{p_2}{p_1}=2:1$$

6. Statement-2 is the explanation of statement-1.

7. Average kinetic energy for diatomic gases

$$K_{av}=\frac{5}{2}kT$$

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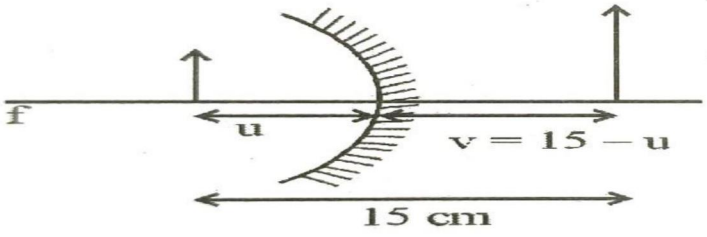
$$\therefore \frac{(K_{av})_H}{(K_{av})_O} = \frac{(27+273)}{(27+273)} = 1$$

8. $$\frac{\Delta P_1}{\Delta P_2} = \frac{r_2}{r_1} = \frac{3}{1}; P = \frac{2T}{r}$$

$$V \propto r^3; M = V \times \rho$$

$$\frac{M_1}{M_2} = \frac{V_1}{V_2} = \frac{r_1^3}{r_2^3} = \frac{1}{27}$$

9.



Since ,image is magnified so,it is a concave mirror.

$$\therefore m = 2 = \frac{-v}{u}$$

$$2 = \frac{-(15-u)}{-u}$$

$$2u = 15 - u$$

$$3u = 15 \Rightarrow u = 5$$

$$v = 15 - u = 15 - 5 = 10cm$$

By mirror formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{10} = \frac{1}{(-5)} + \frac{1-2}{10} = \frac{-1}{10}$$

F=-10cm

10. Dipole mement, $|\vec{p}|$=qd

$=0.01 \times 0.4 \times 10^{-3} = 4 \times 10^{-6}$

And torque, $|\vec{\tau}|$=PE sinθ

$=4 \times 10^{-6} \times 10 \times 10^{-5} \times \sin 30$

$\therefore |\vec{\tau}| = 2 \times 10^{-10}$

11. (A+B).C. For A=1,B=0 and C=1 WILL GIVE 1.

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12. $\frac{K\left(5q^2\right)}{r^2}=\frac{mv^2}{r}.....(1)$

$mvr=\frac{nh}{2\pi}.....(2)$

Solving (1) and (2)

$\Rightarrow V=\frac{10Kq^2.\pi}{nh}$

For 1st orbit $V=\frac{5q^2}{2\varepsilon_0 h}$

13. Given: Distance between one crest and one through=1.5cm

$=\left(2n_1+1\right)\frac{\lambda}{2}$

Distance between two crests=5 m= $n_2\lambda$

$\frac{1.5}{5}=\frac{\left(2n_1+1\right)}{2n_2}\Rightarrow 3n_2=10n_1+5$

Here n_1 and n_2 are integer.

If $n_1=1,\ n_2=5 \qquad \therefore \lambda=1$

$n_1=4,\ n_2=15 \qquad \therefore \lambda=1/3$

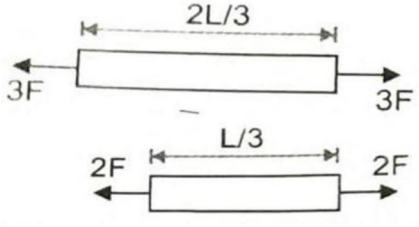
$n_1=7,\ n_2=25 \qquad \therefore \lambda=1/5$

Hence possible wavelengths $\frac{1}{1},\frac{1}{3},\frac{1}{5}$ metre.

14. Lateral displacement, $\Delta x=t\sin i\left(1-\frac{1}{\mu}\right)<t$.

15. As temperature increases, domains disintegrate so ferromagnetism decreases and above Curie's temperature it becomes paramagnet. Also, at high temperature the domain area of ferromagnetic substance decreases.

16. The free body diagram of two parts are shown in figure



Both parts are stretched. Therefore total elongation

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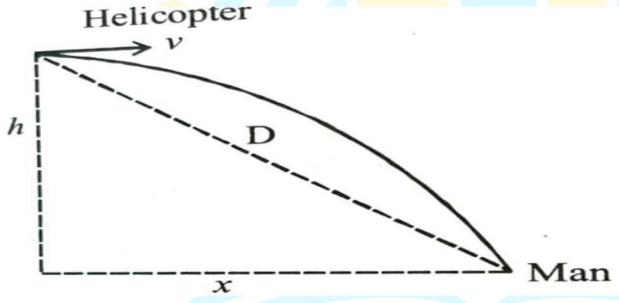
$$\Delta l = \Delta l_1 + \Delta l_2 = \frac{3F\left(\frac{2L}{3}\right)}{AY} + \frac{2F\left(\frac{L}{3}\right)}{AY} = \frac{8FL}{3AY}$$

17. $TV^{\gamma-1} = \text{cons tan t} \Rightarrow T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$$= 300\left\{\left(\frac{27}{8}\right)^{1/3}\right\}^2 = 300\left(\frac{3}{2}\right)^2 = 675\,K$$

$$\Rightarrow \Delta T = 675 - 300 = 375\,K$$

18.



Time of fall of packet

$$t = \sqrt{\frac{2h}{g}}, x = \sqrt{\frac{2h}{g}}v$$

Horizontal range (x)= time× horizontal component of velocity(v)

Required distance

$$D = \sqrt{x^2 + h^2} = \sqrt{\left(\sqrt{\frac{2h}{g}}v\right)^2 + h^2}$$

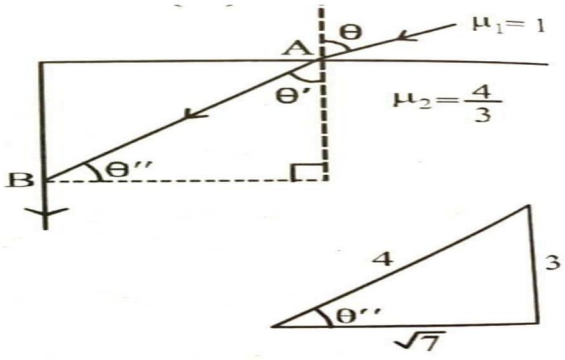
Or, $D = \sqrt{\frac{2hv^2}{g} + h^2}$

19. Relation between acceleration and velocity is:

$$a = v\frac{dv}{dx} = 4\sqrt{x} \times 4 \times \frac{1}{2\sqrt{x}} = 8m/s^2$$

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20.



At maximum angle θ, the ray goes along the surface.

For all angles less than θ, total internal reflection occurs. At point B

$$\frac{4}{3}\times\sin\theta'' = \sin 90^0 \Rightarrow \theta'' = \sin^{-1}\left(\frac{3}{4}\right)$$

$$\theta' = \left(\frac{\pi}{2} - \theta''\right)$$

At point A,

$$1\times\sin\theta = \frac{1}{3}\times\sin\theta'$$

$$\Rightarrow \sin\theta = \frac{4}{3}\times\sin\left(\frac{\pi}{2} - \theta''\right)$$

$$\Rightarrow \sin\theta = \frac{4}{3}\cos\theta''$$

From the figure,

$$\cos\theta'' \frac{\sqrt{7}}{4} \qquad \therefore \sin\theta \frac{4}{3}\times\frac{\sqrt{7}}{4}$$

$$\text{or},\theta = \sin^{-1}\left(\frac{\sqrt{7}}{3}\right)$$

21. $Q_i = Q_{plate}\left(1 - \frac{1}{k}\right)$

$$= KCV\left(1 - \frac{1}{k}\right) = CV(K-1)$$

$$= 90\,p_F \times 20 \times \left(\frac{5}{3} - 1\right) = 1.2n_C$$

22. $P = V_{ms} I_{ms} COS\Phi$

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$$\Rightarrow 400 = \frac{V_{MS}^2}{Z} \times 0.8$$

$$\Rightarrow Z = \frac{250^2}{400} \times 0.8 = 125\Omega$$

$$\cos\Phi = \frac{R}{Z}$$

$$R = Z\cos\Phi = 125 \times 0.8 = 100\Omega$$

Now, $z^2 = x_L^2 + R^2$

$$\Rightarrow 125^2 = x_L^2 + 100^2$$

$$\Rightarrow x_L = 75\Omega$$

At resonance

$$x_L = x_C$$

$$\Rightarrow 75 = \frac{1}{w_C} \Rightarrow C = \frac{1}{2\pi \times 75 \times 50} \Rightarrow C = \frac{400}{3\pi}\mu_F$$

23. In series $\frac{dQ}{dt} = cons\tan t$

$$\Rightarrow \frac{T_1 - T_2}{R} = cons\tan t$$

$$\Rightarrow \frac{450 - T}{\frac{l_2}{K_2 A_2}} = \frac{T - 0}{\frac{l_1}{K_1 A_1}} \Rightarrow \frac{450 - T}{\frac{l_1}{l_2} \cdot \frac{K_2}{K_1} \cdot \frac{A_2}{A_1}} = T$$

$$\Rightarrow \frac{450 - T}{2.9.\frac{1}{2}} = T \Rightarrow 450 - T = 9T$$

$$\Rightarrow 450 = 10T \Rightarrow T = 45^0 C$$

24. Given:Electric field, $\vec{E} = \frac{3}{5}E_0\hat{i} + \frac{4}{5}E_0\hat{J}\frac{N}{C}$

Area (Parallel toy-z plane), $a_a = 0.2\hat{i}m^2$

Area (Parallel tox-z plane), $a_b = 0.3\hat{J}m^2$

Flux, $\phi_a = \vec{E}, \overrightarrow{A_a} = \left(\frac{3}{5}E_0\hat{i} + \frac{4}{5}E_0 j\right).0.2\hat{i} = \frac{12}{50}E_0$

$$\phi_b = \vec{E}, \overrightarrow{A_b} = \left(\frac{3}{5}E_0\hat{i} + \frac{4}{5}E_0 j\right)0.3\hat{i} = \frac{12}{50}E_0$$

$$\therefore \frac{\phi_a}{\phi_b} = \frac{6/50E_0}{12/50E_0} = \frac{1}{2}.So\ a=1$$

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25. Moment of interia of disc, $I=\frac{MR^2}{2}$

$$I_{remaining} = I - I_{removed}$$

$$= \frac{MR^2}{2} - \left[\frac{\frac{m}{4}\left(\frac{R}{2}\right)^2}{2} + \frac{M}{4}\left(\frac{R}{2}\right)^2 \right]$$

$$I_{remaining} = \frac{13}{32}MR^2$$

CHEMISTRY

26. During De-naturation primary structure of proteins does not change

27. Conceptual

28. $MgSO_{4(s)} + xH_2O_{(l)} \rightarrow MgSO_4.xH_2O : \Delta H_1 = -80$ ..... (1)

$MgSO_4.7H_2O_{(s)} + (x-7)H_2O_{(l)} \rightarrow MgSO_4.xH_2O : \Delta H_2 = 12$ .....(2)

(1) – (2) = $\Delta H_{hyd} = -92\ kJ/mol$

29. If maximum volume of O_2 = 100 cm³ , then max.volume of N_2O_5 is $200cm^3$

$2N_2O_5(g) \rightarrow 2N_2O_4(g) + O_2(g)$

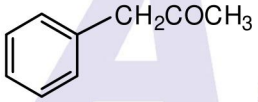
At t = 0: 200 0 0

At t = 500: 200 – x x x/2

$x/2 = 90 \Rightarrow x = 180$

$$k = \frac{2.303}{500}\log\frac{200}{200-180} = \frac{2.303}{500}$$

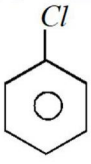
30.



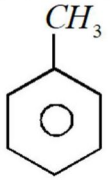
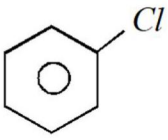
Satisfies all the above reactions.

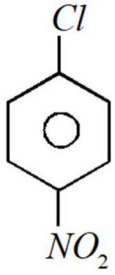
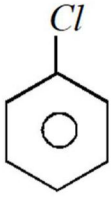
31. Conceptual

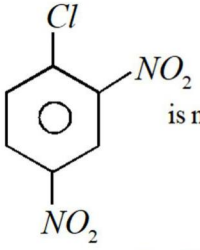
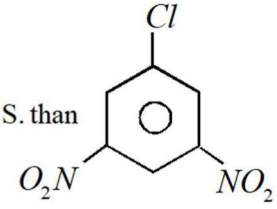
32.


is more reactive towards E.Ar.S. than


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is more reactive towards E.Ar.S. than



is more reactive towards Nu.Ar.S. than



is more reactive towards Nu.Ar.S. than


33. Conceptual

34. $P_T = 0.5\times400 + 0.5\times600 = 500$ mm Hg

$$y_A = \frac{200}{500}; y_B = \frac{300}{500} \Rightarrow 3y_A = 2y_B$$

35. $VO_2, Cr_2O_3, MnO_2, V_2O_5, ZnO -$ amphoteric (5)

Primary valency of 'Cr' is - 3

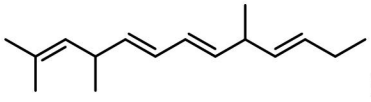
36. Conjugate acid of nitrogen 3 is mole stable because of conjugation

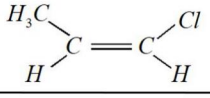
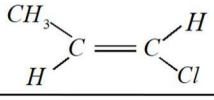
37. $[NiCl_2(PPh_3)_2]$ – 2 unpaired electrons $[V(CO)_6]$ – 1 unpaired electron $[Cr(CN)_6]^{4-}$ – 2 unpaired electrons $[Ni(CO)_4]$ – 0 unpaired electron

38. For ideal solutions $\Delta H_{mix} = 0$, Acetic acid dimerises in benzene solutions shows –ve deviation forms maximum boiling azeotropes, solutions shows +ve deviation forms minimum boiling azeotropes.

39. Conceptual

40. $[Co(CN)_6]^{3-}$, $[Co(C_2O_4)_3]^{3-}$ diamagnetic compounds, magnetic moment is zero for both complexes

41.  is the alkene (A)

42. 


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$\mu_{trans} > \mu_{cis}$: so B.P. of trans > cis isomer

43. Conceptual

44. Conceptual

45. $Fe, Hg \,\&\, V$ are oxidized by NO_3^-

46. $pH = P^{Ka} + \log\dfrac{[salt]}{[Acid]}$; $pH = P^{Ka} = 4.7$ initial pH of the sol.

on adding 0.05 mol of $NaOH$

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

0.01	0.05	0.1
0.05	0	0.1+0.05

Now the $pH = P^{Ka} + \log\dfrac{0.15}{0.05} = 4.7 + 0.48$

$\Delta pH = 0.48$

47. no. of mole of $N_2 = \dfrac{PV}{RT} = \dfrac{1\times 49.2\times 10^{-3}}{0.082\times 300}$

Mass of 'N' = $28\times$ no. of moles

$$\%N = \frac{28\times 49.2\times 10^{-3}}{0.082\times 300}\times\frac{1000}{200}\times 100 = 28\%$$

48. limiting reagent is B

49. $\left[Co(NH_3)_6\right]^{+3}$ & $\left[Co(C_2O_4)_2\right]^{-3}$ are diamagnetic.

Remaining all are paramagnetic

50. $X \rightarrow Benzene$

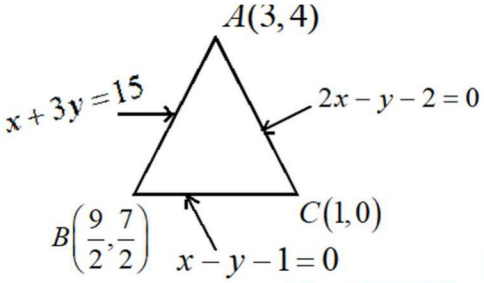
$Y \rightarrow Ph - CO - C_2H_5$

$Z \rightarrow Ph - \underset{\underset{C_2H_5}{|}}{C} = N - NH - (C_6H_3)(NO_2)_2$

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MATHEMATICS

51.



Equation of altitude through A is $x + y = 7$ …(1)

Equation of altitude through B is $x + 2y = \frac{23}{2}$ …(2)

Solving (1) and (2) we will get $H\left(\frac{5}{2}, \frac{9}{2}\right)$

Perpendicular distance of H from line $x + y - 3 = 0$ is

$$\left|\frac{\frac{5}{2} + \frac{9}{2} - 3}{\sqrt{2}}\right| = 2\sqrt{2}$$

52. $2\bar{a} + \bar{b} = -\bar{c}$

$$\left|2\bar{a} + \bar{b}\right|^2 = \left|\bar{c}\right|^2$$

$$4 + 9 + 4\bar{a}.\bar{b} = 16$$

$$4\bar{a}.\bar{b} = 3$$

$$\left|\bar{a} + 2\bar{b}\right|^2 = 1 + 36 + 4\bar{a}.\bar{b} = 40$$

$$\left|\bar{a} + 2\bar{b}\right| = 2\sqrt{10}$$

53. $|x - 1| \le 2$

$-2 \le x - 1 \le 2$

$-1 \le x \le 3$

$A = [-1, 3]$

$2 \le |y + 1| < 6$

$2 \le y + 1 < 6$ *or* $-6 < y + 1 \le -2$

$1 \le y < 5$ *or* $-7 < y \le -3$

$B = (-7, -3] \cup [1, 5)$

$A \cap B = [1, 3]$

$A - B = A - A \cap B$

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$= [-1,1)$

54. $y = \dfrac{2x^2 - 5x + 3}{4x - \lambda}$

Cross multiply

$2x^2 - x(4y+5) + (\lambda y + 3) = 0,$

$D \geq 0 \because x \in R$

$(4y+5)^2 \geq 8(\lambda y + 3)$

$16y^2 + 8y(5-\lambda) + 1 \geq 0 \quad \forall\, y \in R.$

$D < 0$

$64(5-\lambda)^2 < 64$

$\lambda^2 - 10\lambda + 24 \leq 0$

$(\lambda - 4)(\lambda - 6) < 0$

$\lambda \in (4,6)$

55. Required probability $= \dfrac{\frac{2}{10} \times \frac{2}{3}}{\frac{2}{10} \times \frac{2}{3} + \frac{8}{10} \times \frac{1}{2}} = \dfrac{1}{4}$

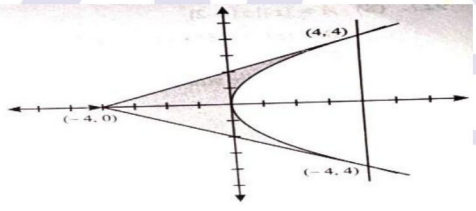
56. $P(x_i) = m.x_i + c, i = 0,1,2,3,4$

$\sum_{i=0}^{4} p(x_i) = 1 \Rightarrow 2m + c = \dfrac{1}{5}$

$\sum_{i=0}^{4} x_i p(x_i) = \dfrac{5}{2} \Rightarrow 3m + c = \dfrac{1}{4} \Rightarrow m = \dfrac{1}{20}, c = \dfrac{1}{10}$

$\Rightarrow \alpha = \sum_{i=0}^{4} x_i^2 p(x_i) - \left(\sum_{i=0}^{4} x_i p(x_i)\right)^2 = 8 - \dfrac{25}{4} = \dfrac{7}{4} \Rightarrow 24\alpha = 42$

57.



Area of the shaded region is

$\dfrac{1}{2} \times 8 \times 8 - 2\int_0^4 2\sqrt{x}\, dx$

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$$= 32 - 4 \times \frac{2}{3} x^{3/2} \Big|_0^4$$

$$= 32 - \frac{8}{3} \times 8$$

$$= \frac{1}{3} \times 32$$

$$\Rightarrow \frac{32}{3} = \Delta$$

$$\therefore [\Delta] = 10$$

58. If $\lim_{x \to 0} \frac{f(x)}{x^3} = 1$

then $f(x) = x^3 + bx^4$

$$f'(x) = 3x^2 + 4bx^3 = x^2(3 + 4bx)$$

$$f'\left(\frac{1}{8}\right) = 0$$

$$3 + \frac{b}{2} = 0 \; b = -6$$

$$\lim_{x \to 0} \frac{f(x) - x^3}{x^4} = b = -6$$

59. $3\sin\theta - 4\sin^3\theta = 4\sin\theta.\sin 2\theta.\sin 4\theta$

$3 - 4\sin^2\theta = 4\sin 2\theta \sin 4\theta$ **OR** $\sin\theta = 0$

$3 - 2(1 - \cos 2\theta) = 2.(\cos 2\theta - \cos 6\theta)$ $\theta = 0, \pi$

$1 + 2\cos 2\theta = 2\cos 2\theta - 2\cos 6\theta$

$\cos 6\theta = -\frac{1}{2} = \frac{\cos 2\pi}{3}, \; 6\theta \in [0, 6\pi]$

2 solutions in $[0, 2\pi]$

6 solutions in $[0, 6\pi]$

Total = $2 + 6 = 8$ solutions

60. $T_6 + T_{19} = T_{10} + T_{15} = T_{12} + T_{13} = x$

$3x = 600$

$x = 200 \quad = T_1 + T_{24}$

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$$S_{24}=\frac{24}{2}\left(T_1+T_{24}\right)=2400$$

61. $\frac{2+3z+4z^2}{2-3z+4z^2}$ is a real number

$$\Rightarrow 1+\frac{6}{\frac{2}{z}-3+4z}\in R\Rightarrow 2z+\frac{1}{z}\in R\Rightarrow 2z+\frac{1}{z}=2\bar{z}+1/\bar{z}$$

$=(2z\bar{z}-1)(z-\bar{z})=0$

$$\Rightarrow |Z|^2=\frac{1}{2}=0.50$$

62. By the properties of an ellipse, $|PF_1|+|PF_2|=2a_1$,and by the properties of a hyperbola, $|PF_1|-|PF_2|=2a_2$. These Equations combine to $|PF_1|=a_1+a_2$ and $|PF_2|=a_1-a_2$. Also by the properties of ellipse and hyperbola, $|F_1F_2|=2a_1e_1=2a_2e_2$. since $|F_1F_2|=2|PF_2|, 2a_1e_1=2(a_1-a_2)$ .Substituting $a_2=\frac{a_1e_1}{e_2}$ and solving gives $e_2=\frac{e_1}{1-e_1}$.

For a hyperbola, $e_2>1$, which means $\frac{e_1}{1-e_1}>1$, which solves to $e_1>\frac{1}{2}$.

For an ellipse, $0<e_1<1$. Therefore $\frac{1}{2}<e_1<1$.

Therefore, D=$e_2-e_1=\frac{e_1}{1-e_1}-e_1=\frac{e_1^2}{1-e_1}$. For $\frac{1}{2}<e_1<1$,the range of D is $\left(\frac{1}{2},\infty\right)$.

63. Given, $\frac{xdy}{dx}-2y=x^4y^2$ or $\frac{dy}{dx}-\frac{2}{x}y=x^3y^2$,Now dividing by y^2,we get

$$\frac{1}{y^2}\frac{dy}{dx}-\frac{2}{y}\frac{1}{x}=x^3$$

This is a Bernouli's differential equation, substituting $\frac{-2}{y}=t$,we get

$$\Rightarrow \frac{2}{y^2}\frac{dy}{dx}=\frac{dt}{dx}.So, equation(i) becomes$$

$$\Rightarrow \frac{1}{2}\frac{dt}{dx}+\frac{t}{x}=x^3\Rightarrow \frac{dt}{dx}+\frac{2}{x}t=2x^3$$

I.F.=$e^{\int\frac{2}{x}dx}=e^{2\ln x}=x^2$

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, general solution is given by

$$x^2t=\frac{2x^6}{6}+C\Rightarrow\frac{-x^2.2}{y}=\frac{x^6}{3}+C\Rightarrow\frac{-2}{y}=\frac{x^4}{3}+\frac{c}{x^2}$$

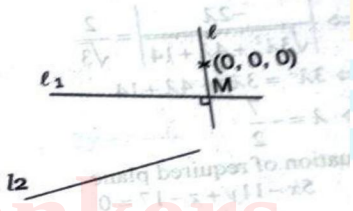
If x=1,y=-6, $\Rightarrow C=0$

$$\therefore\quad\frac{-2}{y}=\frac{x^4}{3}\Rightarrow y=\frac{-6}{x^4}\,i.e.,\,f(x)=\frac{-6}{x^4}$$

Now, $\frac{dy}{dx}=24x^{-5}=\frac{24}{x^5}.$

Hence, $\left.\frac{dy}{dx}\right]_{x=3^{1/5}}=\frac{24}{3}=8$

64. The common perpendicular is along $\begin{vmatrix}\hat{i} & j & k\\ 1 & 2 & 2\\ 2 & 2 & 1\end{vmatrix}$=2î+3ĵ-2ǩ



Let M=$(2\lambda,3\lambda,2\lambda)$

So, $\frac{2\lambda-3}{1}=\frac{-3\lambda+1}{2}=\frac{2\lambda-4}{2}\Rightarrow\lambda=1$

So, M=(2,-3,2)

Let the required point be P

Given that PM=$\sqrt{17}$

$\Rightarrow(3+2s-2)^2+(3+2s+3)^2+(2+s-2)^2=17$

$\Rightarrow 9s^2+28s+20=0$

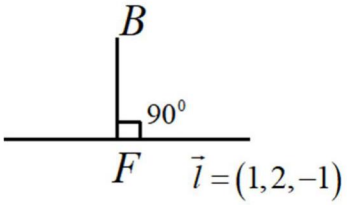
$\Rightarrow S=-2,-\frac{10}{9}$

65. $\frac{{}^{10}C_1}{4}.{}^{7-3+1}C_3$

$=\frac{5}{2}\times{}^5C_3=25$

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66. $\frac{x-2}{1}=\frac{y-1}{2}=\frac{z-1}{-1}=\lambda$



Let $F(\lambda+2,2\lambda+1,1-\lambda)$

$\overrightarrow{BF}(\lambda,2\lambda-2,2-\lambda)$ $\quad\overrightarrow{BF}.\vec{l}=0$

$(\lambda,2\lambda-2,2-\lambda)(1,2,-1)$

$\lambda+4\lambda-4+\lambda-2=0$

$\lambda=1$

$F(3,3,0)$ $\quad\overrightarrow{BF}=(1,0,1)$

Area of $\Delta ABF=\frac{1}{2}\left|\overrightarrow{AB}\times\overrightarrow{BF}\right|$

$=\frac{1}{2}\left|\left(\hat{i}+\hat{j}-\hat{k}\right)\times\left(\hat{i}+\hat{k}\right)\right|$

$=\frac{1}{2}\left|\hat{i}-2\hat{j}-\hat{k}\right|$

$S=\frac{\sqrt{6}}{2}$

$2S^2=2\times\frac{6}{4}=3$

67. $\left|2^x-1\right|+\left|2^x-4\right|=3$

Case-I: x ∈ (-∞,0)

$-2^x+1-2^x+4=3\Rightarrow-2.2^x=-2\Rightarrow2^x=1\Rightarrow x=0$ (rejected)

Case-Π: $x\in[0,2]$

$-2^x-1-2^x+4=3\Rightarrow3=3$

Case-ΠI: $x\in(2,\infty)$

$-2^x-1+2^x-4=3\Rightarrow2.2^x=8=4\Rightarrow x=2$

68. Conceptual

69. $D=\begin{vmatrix}2&1&1\\1&-1&1\\1&1&a\end{vmatrix}=0$

$\Rightarrow$2(-a-1)-1(a-1)+1+1=1-3a=0

$\Rightarrow a=\frac{1}{3}$

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$$D_3=\begin{vmatrix}2 & 1 & 5\\ 1 & -1 & 3\\ 1 & 1 & b\end{vmatrix}=2(-b-3)-1(b-3)+5(1+1)$$

$$=7-3b\neq 0\Rightarrow b\neq\frac{7}{3}$$

Hence, for a=$\frac{1}{3}, b\neq\frac{7}{3}$, system has no solutions

70. CONCEPTUAL

71. $\cos(\tan x)-\cos x=2\sin\left(\frac{x+\tan x}{2}\right)\sin\left(\frac{x-\tan x}{2}\right)$

or $\lim\limits_{x\to 0}\dfrac{\cos(\tan x)-\cos x}{x^4}$

$$=\lim_{x\to 0}\frac{2\sin\left(\frac{x+\tan x}{2}\right)\sin\left(\frac{x-\tan x}{2}\right)}{x^4}$$

$$=\lim_{x\to 0}\frac{2\sin\left(\frac{x+\tan x}{2}\right)\sin\left(\frac{x-\tan x}{2}\right)}{x^4\left(\frac{x+\tan x}{2}\right)\left(\frac{x-\tan x}{2}\right)}\left(\frac{x^2-\tan^2 x}{4}\right)$$

$$=\frac{1}{2}\lim_{x\to 0}\frac{x^2-\tan^2 x}{x^4}$$

$$=\frac{1}{2}\lim_{x\to 0}\frac{x^2-\left(x+\frac{x^3}{3}+\frac{2}{15}x^5+\ldots\right)^2}{x^4}$$

$$=\frac{1}{2}\lim_{x\to 0}\frac{1}{x^2}\left(1-\left(1+\frac{x^2}{3}+\frac{2}{15}x^4+\ldots\right)\right)=-\frac{1}{3}$$

72. $I=\int\dfrac{\sin\theta.\sin 2\theta\left(\sin^6\theta+\sin^4\theta+\sin^2\theta\right)\sqrt{2\sin^4\theta+3\sin^2\theta+6}}{1-\cos 2\theta}d\theta$

$$\Rightarrow I=\int\frac{\sin\theta.2\sin\theta\cos\theta.\sin^2\theta\left(\sin^4\theta+\sin^2\theta+1\right)\left(2\sin^4\theta+3\sin^2\theta+6\right)^{1/2}}{2\sin^2\theta}d\theta$$

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$$= \int \sin^2\theta.\cos\theta\left(\sin^4\theta+\sin^2\theta+1\right)\left(2\sin^4\theta+3\sin^2\theta+6\right)^{1/2} d\theta$$

Let $\sin\theta = t \Rightarrow \cos\theta\, d\theta = dt$

$$\therefore I = \int t^2\left(t^4+t^2+1\right)\left(2t^4+3t^2+6\right)^{1/2} dt$$

$$= \int \left(t^5+t^3+t\right)\left(2t^6+3t^4+6t^2\right)^{1/2} dt$$

$$= \frac{1}{12}\int \left(2t^6+3t^4+6t^2\right)^{1/2}\left(12t^5+12t^3+12t\right)dt$$

$$= \frac{\left(2t^6+3t^4+6t^2\right)^{3/2}}{12\times\frac{3}{2}} + C$$

$$= \frac{\left(2t^6+3t^4+6t^2\right)^{3/2}}{18} + C \text{, where } t = \sin\theta$$

73. Required probability is $\frac{1}{5} = \frac{m}{n}$, then m^2+n^2 is $1+25=26$

74. Let the semi-major axis, semi-minor axis, and eccentricity of the ellipse be $a, b,$ and $e,$ respectively, and those of hyperbola be $a', b',$ and $e',$ respectively.

Given that $a - a' = 4$ (1)

$2ae = 2\sqrt{13}$ or $ae = \sqrt{13}$ (2)

$2a'e' = 2\sqrt{13}$ or $a'e' = \sqrt{13}$ (3)

and $\frac{e}{e'} = \frac{3}{7}$ (4)

From (2), $a = \frac{\sqrt{13}}{e}$ (5)

From (3), $a' = \frac{\sqrt{13}}{e'}$ (6)

Putting in (1), we get

$$\frac{\sqrt{13}}{e} - \frac{\sqrt{13}}{e'} = 4$$

or $\sqrt{13}\left(e'-e\right) = 4ee'$

Putting $e = \frac{3e'}{7}$, we get

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$$\sqrt{13}\left(e'-\frac{3}{7}e;\right)=4\times\frac{3}{7}(e')^2$$

$$\therefore e'=\frac{\sqrt{13}}{3} and\ e=\frac{3}{7}e'=\frac{\sqrt{13}}{7} \qquad a=\frac{\sqrt{13}}{e}=\frac{\sqrt{13}}{\sqrt{13}}\times 7=7$$

$$a'=\frac{\sqrt{13}}{e'}=\frac{\sqrt{13}}{\sqrt{13}}\times 3=3 \qquad b^2=a^2\left(1-e^2\right)=49\left(1-\frac{13}{49}\right)=36$$

$$(b')^2=(a')^2\left\{(e')^2-1\right\}=9\left(\frac{13}{9}-1\right)=4$$

The equation of ellipse is

$$\frac{x^2}{49}+\frac{y^2}{36}=1$$

and the equation of hyperbola is

$$\frac{x^2}{9}-\frac{y^2}{4}=1$$

75. $S=C_0-2^2C_1+3^2C_2-\ldots+(-1)^n(n+1)^2C_n$

$$T_r=(-1)^r r^2\,{}^nC_r$$

$$=(-1)^r r\left(r\,{}^nC_r\right)$$

$$=(-1)^r r\left(n^{n-1}C_{r-1}\right)$$

$$=n(-1)^r\left((r-1)+1\right)\left({}^{n-1}C_{r-1}\right)$$

$$=n(-1)^r\left((r-1)^{n-1}C_{r-1}+{}^{n-1}C_{r-1}\right) \qquad =n(-1)^r\left((n-1)^{n-2}C_{r-2}+{}^{n-1}C_{r-1}\right)$$

$$=n(n-1)^{n-2}C_{r-2}(-1)^{r-2}-n^{n-1}C_{r-1}(-1)^{r-1}$$

$$\sum_{r=0}^{n}T_r$$

$$=n(n-1)(1-1)^{n-2}-n(1-1)^{n-1}=0$$

Similarly $\sum(-1)^r r.\,{}^nC_r=0$

$$\sum(-1)^r\,{}^nC_r=0$$

$$S=0$$

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